Chunxu Han

Peder Skrams Gade 16A, København K, Denmark s220311@dtu.dk +45 52 73 06 85 in chunxu-han
chunxubioinfor

Relevant Experience

Machine Learning for High-Dimensional Omics Data
My master's thesis at Novo Nordisk focused on developing and benchmarking computational methods to evaluate batch-to-batch variability in single-cell RNA-seq data — a high-dimensional setting where model evaluation and generalizability across datasets are central challenges. I designed quantitative metrics from scratch, implemented a systematic benchmarking framework comparing multiple computational approaches, and explored machine learning methods to correlate transcriptomic variability with functional outcomes. This work resulted in **scBatch**, a Python package built around reproducibility and modular design.

Reproducible Pipelines and Research Software Development
Over the past two years, I contributed to the development and maintenance of IsoformSwitchAnalyzeR, an R package for isoform-level analysis of long-read and single-cell sequencing data, which I helped bring to a published preprint. Across both roles, I built reproducible bioinformatics pipelines on HPC systems using Snakemake and Git, and worked extensively with large-scale omics datasets — including curation and re-quantification of 40,000+ proteomics samples. I am used to designing workflows that are not just functional, but transferable and well-documented.

Collaborative Research Across Academic and Industrial Settings
I have worked at the intersection of methods development and translational application — first in an academic isoform analysis group at DTU HealthTech, then in an industry-facing R&D setting at Novo Nordisk, collaborating with cross-functional CMC and R&D teams. I enjoy contributing to research that ultimately has a clinical or patient-relevant angle, and I find it motivating to work in environments where rigorous computational work connects to real biological questions.

Education

MSc in Bioinformatics and Systems Biology, Technical University of Denmark — Lyngby, Denmark, Sept 2023 – March 2026

- Completed **Machine Learning** and **Deep Learning** courses alongside specialised bioinformatics training, building a strong foundation for developing and evaluating ML models in biological contexts.

BSc in Biotechnology, Beijing Normal University — Beijing, China, Sept 2017 – June 2021

- Foundation in molecular biology, cell biology, and bioinformatics, with hands-on laboratory training.

Work Experience

Master Student, Novo Nordisk — Målev, Denmark, Sep 2025 – Mar 2026

Thesis: developing computational frameworks to evaluate batch-to-batch variability in single-cell RNA-seq data for cell therapy quality control.

- **Benchmarking and Methodology Development:** Designed and implemented a systematic benchmarking framework comparing multiple computational tools for assessing batch comparability in cell-type composition and transcriptional profiles.
- **Machine Learning:** Explored ML approaches to correlate transcriptomic variability with functional outcomes, contributing to predictive quality assessment in a GMP-adjacent setting.
- **Software:** Developed **scBatch**, a Python package providing a structured and reproducible framework for batch variability assessment, with a focus on modular

design and clear interfaces.

- **Collaboration:** Worked with cross-functional R&D and CMC teams to ensure biological interpretability and industrial relevance of results.

Bioinformatics Research Assistant, DTU HealthTech — Lyngby, Denmark, Oct 2023 – July 2025

Two years in the Isoform Analysis Group, contributing to both software development and large-scale data analysis.

- **Software Development:** Contributed to developing and updating IsoformSwitchAnalyzeR v2 (R), supporting isoform-level analysis of long-read and single-cell RNA-seq data. The package is now available as a preprint on bioRxiv.
- **Bioinformatics Pipelines:** Built and optimised reproducible pipelines on HPC infrastructure using Snakemake and Git; curated and re-quantified 40,000+ proteomics samples from GPMdb following FAIR data principles.
- **Collaboration:** Supported PhD students, postdocs, and master's students with data preprocessing, pipeline troubleshooting, and result interpretation.

Consulting Intern, Illuminera, IQVIA — Shanghai, China, Feb 2023 – May 2023

- Conducted quantitative market and competitor analysis for pharmaceutical clients; delivered data-driven strategic reports using Power BI.

QC Intern, SINOVAC BIOTECH LTD. — Beijing, China, Oct 2020 – Nov 2020

- Worked in a GMP-regulated Quality Control lab; techniques include HP-LC, ELISA, SDS-PAGE, Lowry Protein Assay.

Publications

- **Han, C.**, Gilis, J., Iriondo Delgado, E., Clement, L., & Vitting-Seerup, K.
  IsoformSwitchAnalyzeR v2: Analysis of Functional Isoform Changes in Long-read and Single-cell Sequencing Data.
  bioRxiv (2025). doi:10.64898/2025.12.08.693027
- Mikkelsen, J. M., **Han, C.**, Kanakoglou, D. S., Tiberi, S., & Vitting-Seerup, K.
  Beyond the Gene in Genetics: How Isoform-Resolved Analysis Empowers the Study of Both Common and Rare Genetic Variation.
  medRxiv (2025). doi:10.1101/2025.03.31.25324931

Skills

Programming: Python, R, Bash, SQL, Snakemake, Docker

Bioinformatics: Single-cell RNA-seq, isoform analysis, long-read sequencing, HPC pipelines, batch effect methods

Machine Learning: Supervised/unsupervised learning, deep learning, model evaluation and benchmarking, foundation models for omics

Languages: English (proficient), Chinese (native), Danish (basic)

Hobbies: I've been playing **badminton** since an early age. I also enjoy **traveling** around Europe, **hiking**, and **sailing** — last summer we sailed to Ven and Hallands Väderö, which was a completely new kind of freedom.